Quantification: 543_3
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	11
interior lacuna count	10
frame-edge lacuna count	1

Interior lacunae only: mean, median and sample SD (n = 10)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2322.1	2188.5	707.56	px^2
major_axis_length_px	lacuna shape	92.32	87.9273	32.8554	px
minor_axis_length_px	lacuna shape	36.1279	35.0553	10.7489	px
aspect_ratio	lacuna shape	2.7516	2.3867	1.3263	unitless
eccentricity	lacuna shape	0.8916	0.9075	0.0803	unitless
solidity	lacuna shape	0.8685	0.9204	0.1053	unitless
roots_count	headline	3	2	2.6667	count
ring_length_r30_px	headline	283.2	265	121.654	px
ring_length_r60_px	reported	841.7	834	255.132	px
owned_length_px	ownership-dependent	2392.45	1743.56	2469.66	px
edge_count	ownership-dependent	107.7	79.5	111.934	count
mean_edge_length_px	ownership-dependent	20.8378	22.1141	9.8415	px
perimeter_px	normalised	254.343	240.007	75.7536	px
ring_area_r30_px2	normalised	9495.2	8903.5	1859.58	px^2
ring_area_r60_px2	normalised	23331.5	22866	2925.24	px^2
in_frame_fraction_r30	normalised	0.9969	1	0.0099	unitless
in_frame_fraction_r60	normalised	0.9761	1	0.0633	unitless
ring_density_r30	normalised	0.0290706	0.0290657	0.00900672	px^-1
ring_density_r60	normalised	0.0355679	0.0359475	0.00726084	px^-1
roots_per_100px_perimeter	normalised	1.0675	0.9615	0.7454	per 100 px
ring_attached_length_r30_px	network v2	224.3	212	137.845	px
ring_attached_length_r60_px	network v2	566.8	506	325.281	px
ring_length_w_r30_px	network v2	319.454	298.315	141.071	px
ring_length_w_r60_px	network v2	956.504	955.229	290.097	px
sholl_crossings_r10	network v2	6.9	6.5	3.9567	count
sholl_crossings_r20	network v2	10.7	11.5	3.653	count
sholl_crossings_r30	network v2	11.8	11	3.7653	count
ring_width_mean_r30_px	width	7.238	7.3845	0.4152	px

Per field
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measure	value	units
analysed_area_px2	1.02482e+06	px^2
total_skeleton_length_px	35628	px
canalicular_length_density_per_px	0.034765	px^-1
junction_count	869	count
junction_density_per_px2	0.00084795	px^-2
lacuna_count	11	count
lacunae_per_px2	1.073e-05	px^-2
skeleton_component_count	170	count

mean_component_length_px	209.577	px
median_component_length_px	17	px
field_length_density_w_per_px	0.0395898	px^-1
field_density_without_flagged_per_px	0.034765	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	224	px
unattached_component_count	15	count
canalicular_length_density_attached_per_px	0.034765	px^-1
field_length_density_with_unattached_per_px	0.0349836	px^-1

Canalicular width, the new measure

measure	value	units
width_mean_px	7.7028	px
width_median_px	8	px
width_p10_px	4.4721	px
width_p90_px	10	px
width_pixels_used	35628	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2174	80.8669	35.2412	2.2947	0.9	0.9518	0.4109	79.3965	270.898	False
2	3447	90.3677	63.9486	1.4131	0.7066	0.674	0.4223	197.059	916.902	False
3	1883	69.2617	34.8694	1.9863	0.864	0.9448	0.5742	250.833	340.47	False
4	3536	181.867	30.1414	6.0338	0.9862	0.7148	0.0738	327.046	42.4075	False
5	2052	69.2015	39.4228	1.7554	0.8219	0.9112	0.0622	388.671	369.939	False
6	2266	88.1613	35.5691	2.4786	0.915	0.9021	0.1324	473.05	745.191	False
7	1344	87.6934	23.0299	3.8078	0.9649	0.7764	0.2206	545.19	876.866	False
8	1683	71.0123	31.3735	2.2634	0.8971	0.9355	-0.0518	634.68	198.315	False
9	2203	91.2092	31.4053	2.9043	0.9389	0.9295	0.0832	734.301	635.44	False
10	2633	93.5592	36.2778	2.579	0.9218	0.9454	0.0457	780.576	481.495	False
11	531	28.7881	24.8252	1.1596	0.5063	0.9332	0.5942	1012.28	184.834	True

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	2	252	825	184.338	5	36.8676	6.6122
2	9	518	1368	4885.21	223	21.9068	7.3395
3	0	166	570	0	0	0	7.4296
4	5	390	907	6468.61	299	21.6341	7.5552
5	2	265	619	537.073	22	24.4124	7.098
6	5	303	912	2767.86	124	22.3214	7.818
7	1	182	843	2513.11	135	18.6156	6.7112
8	1	108	503	10.8284	1	10.8284	6.7729
9	3	383	1083	5583.47	233	23.9634	7.5455
10	2	265	787	974.009	35	27.8288	7.498
11	0	79	410	0	0	0	6.0343

Normalised per lacuna

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lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	210.066	8595	21715	1	0.9616	0.0293194	0.0379922	0.9521
2	363.446	12097	29071	1	1	0.0428205	0.0470572	2.4763
3	184.794	7941	19515	1	1	0.0209042	0.0292083	0
4	415.99	13536	26753	0.9687	0.7992	0.0288121	0.0339027	1.202
5	205.965	8333	20507	1	1	0.0318013	0.0301848	0.971
6	241.907	9380	23807	1	1	0.0323028	0.0383081	2.0669
7	248.25	8529	21925	1	1	0.021339	0.0384493	0.4028
8	191.965	7943	21525	1	1	0.0135969	0.0233682	0.5209
9	238.108	9212	24077	1	1	0.0415762	0.0449807	1.2599
10	242.936	9386	24420	1	1	0.0282335	0.0322277	0.8233
11	95.3909	3354	9525	0.6102	0.5725	0.023554	0.0430446	0

Network v2 per lacuna

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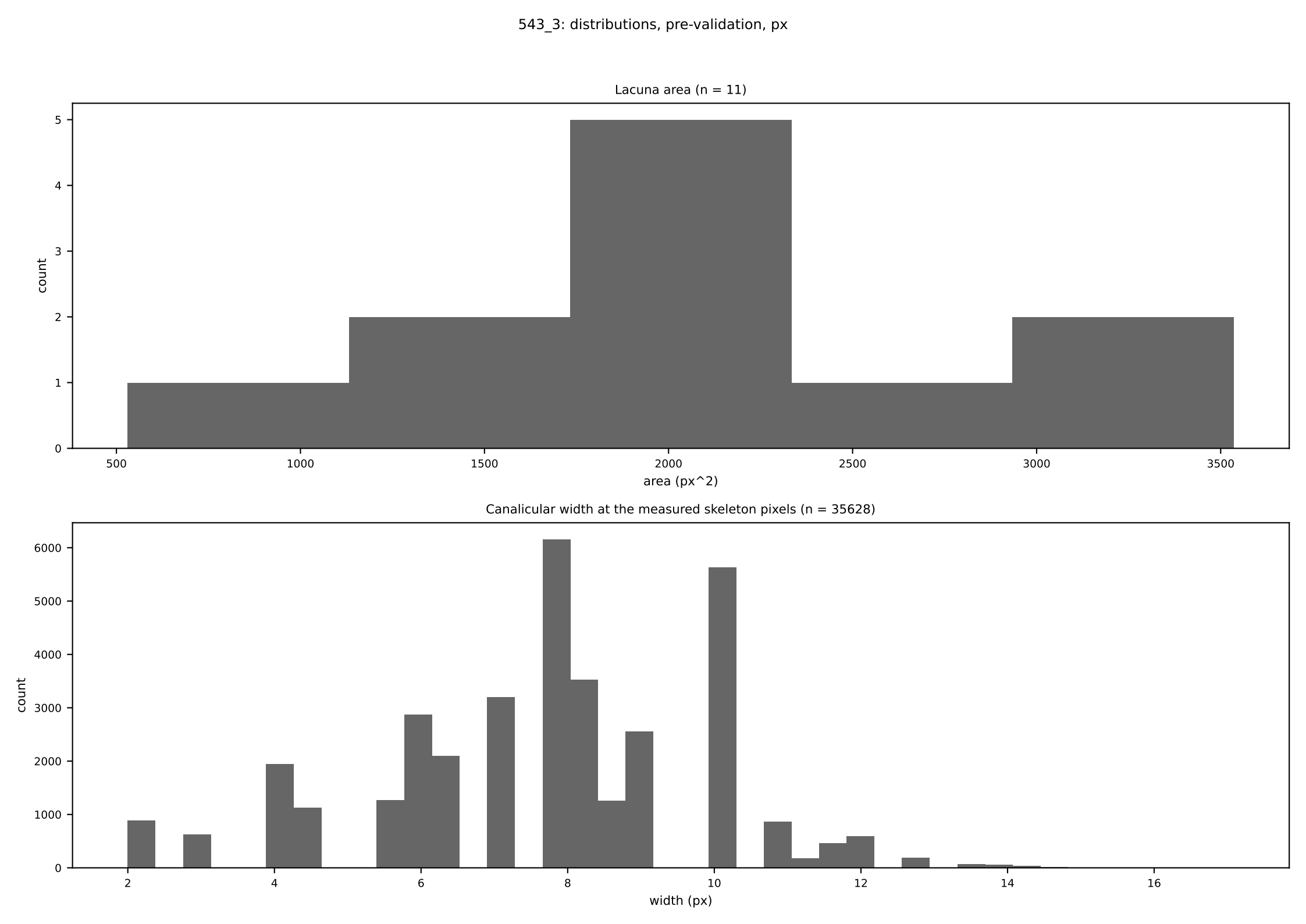
lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	112	318	291.321	958.691	4	12	15
2	469	1167	595.115	1556.15	13	16	19
3	116	384	181.64	645.973	4	7	9
4	379	844	436.191	1029.65	13	13	9
5	238	546	293.865	712.027	7	11	11
6	269	661	338.936	1036.68	7	11	10
7	77	226	201.681	951.767	2	7	11
8	64	153	117.083	561.617	3	4	6
9	333	903	435.948	1220.55	10	14	15
10	186	466	302.764	891.938	6	12	13
11	6	6	84.3553	467.061	1	3	7


543_3: distributions, pre-validation, px
Lacuna area (n = 11)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 35628)
count
width (px)